

FGFR3 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes FGFR3 gene fusions from the msk_impact_50k_2026 study, covering 152 fusion events. 74/152 fusions (48.7%) were in-frame. 142/152 fusions (93.4%) retained the Protein tyrosine and serine/threonine kinase domain. Domain retention was tested with Fisher's exact test ($p=0.0194824$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for FGFR3.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 152 analyzed fusion events, identifying 15 distinct fusion partner genes. The most frequent partner was TACC3, observed in 131 events.

Domain retention

Protein tyrosine and serine/threonine kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 73/74 (98.6%) of in-frame fusions retained the domain, $p=0.0194824$ (statistically significant at $\alpha=0.05$). The Protein tyrosine and serine/threonine kinase appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

Domain disruption

Disruption of the Immunoglobulin domain and Immunoglobulin I-set domain was tested with Fisher's exact test comparing in-frame fusions against all others; 1/74 (1.4%) of in-frame fusions disrupted the domain, $p=0.934975$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.990099.

Cutpoint detection

Cutpoint detection scanned 151 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 677 aa (permutation-corrected $p=0.00990099$, statistically significant at $\alpha=0.05$). This is -71 aa from the nearest configured domain boundary at 748 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=142$) and "not_retained" ($n=9$). For Frame_status, retained: 73/74 (98.6%, 95% CI 92.7%-99.8%); not_retained: 1/2 (50.0%, 95% CI 9.5%-90.5%). Welch's t-test comparing Tumor_variant_count between groups gave means of 109.852 vs 37.4444, $p=0.00694554$ (statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (15 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: FGFR3 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TACC3	131	0.8618421052631579	0.20043213737826426	0.00043213737826425783	0.8507823563952885	0.559399154836743	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.4923043623421339	1
TNIP2	3	0.019736842105263157	0.20043213737826426	0.00043213737826425783	0.8763197586726998	0.559399154836743	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.24350339373637714	2
CCDC149	2	0.013157894736842105	0.20043213737826426	0.00043213737826425783	0.8778280542986425	0.559399154836743	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.24175595386974225	3
JAKMIP1	2	0.013157894736842105	0.20043213737826426	0.00043213737826425783	0.8778280542986425	0.559399154836743	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.24175595386974225	4
LPCAT2	2	0.013157894736842105	0.20043213737826426	0.00043213737826425783	0.8770739064856712	0.559399154836743	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.24164283169779655	5
PTBP1	1	0.006578947368421052	0.20043213737826426	0.00043213737826425783	0.8778280542986425	0.559399154836743	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.23978226965921592	6
CTBP1	1	0.006578947368421052	0.20043213737826426	0.00043213737826425783	0.8763197586726998	0.559399154836743	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.23955602531532452	7
MAEA	1	0.006578947368421052	0.20043213737826426	0.00043213737826425783	0.8506787330316742	0.559399154836743	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.23570987146917066	8
ROBO1	1	0.006578947368421052	0.20043213737826426	0.00043213737826425783	0.8159879336349924	0.559399154836743	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2305062515596684	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
NSD2	3	0.019736842105263157	0.20043213737826426	0.00043213737826425783	0.6561085972850679	0.559399154836743	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.21047171952823235	10
RBM20	1	0.006578947368421052	0.20043213737826426	0.00043213737826425783	0.2413273001508296	0.559399154836743	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.14430715653704396	11
BABAM2	1	0.006578947368421052	0.20043213737826426	0.00043213737826425783	0.20965309200603321	0.559399154836743	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.13955602531532452	12
SLIT2	1	0.006578947368421052	0.20043213737826426	0.00043213737826425783	None	0.559399154836743	['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']	0.12718595472284652	13
GRK4	1	0.006578947368421052	0.20043213737826426	0.00043213737826425783	0.0015082956259426794	0.559399154836743	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.10833430585831094	14
DGKQ	1	0.006578947368421052	0.20043213737826426	0.00043213737826425783	0.0	0.559399154836743	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.10810806151441954	15

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
14	1	141	0	9	None	1.0	-0.0
15	2	140	0	9	None	1.0	-0.0
42	2	140	1	8	0.11428571428571428	0.1694208631494733	0.7710331099865381
127	3	139	1	8	0.17266187050359713	0.21992905390389725	0.6577173940286984
153	4	138	1	8	0.2318840579710145	0.26768849958325047	0.5723702865449334
174	4	138	2	7	0.10144927536231885	0.04197605082466328	1.3769984229650098
188	4	138	3	6	0.057971014492753624	0.004616748961172982	2.33566373986112
311	4	138	4	5	0.036231884057971016	0.00037923555536042514	3.421090951996021
376	4	138	5	4	0.02318840579710145	2.3640024802728035e-05	4.626352072135864
648	4	138	6	3	0.014492753623188406	1.1022799082261042e-06	5.957708108603845

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
653	4	138	7	2	0.008281573498964804	3.6665492410410534e-08	7.4357424781390336
677	4	138	8	1	0.0036231884057971015	7.825580002902196e-10	9.106483464295888
753	5	137	8	1	0.004562043795620438	2.021680915172072e-09	8.694287388663584
758	52	90	8	1	0.07222222222222222	0.002807318578004277	2.5517083002909002
759	88	54	8	1	0.2037037037037037	0.15623087834924132	0.806233125571681
760	89	53	8	1	0.2099056603773585	0.15840511374693833	0.8002308023157874
762	91	51	8	1	0.22303921568627452	0.16500164368454323	0.7825117294856598
763	93	49	8	1	0.2372448979591837	0.27269821801700755	0.5643177000044979
764	95	47	8	1	0.2526595744680851	0.2731204045238703	0.5636458529869772
766	96	46	8	1	0.2608695652173913	0.27473748927465147	0.5610820749288234
767	97	45	8	1	0.26944444444444443	0.27730934214956404	0.5570354994244366
768	100	42	8	1	0.2976190476190476	0.4468103647595553	0.34987676101374454
769	101	41	8	1	0.3079268292682927	0.4455794616522395	0.3510748355557522
770	105	37	8	1	0.3547297297297297	0.45054685984922543	0.34626003284121815
771	108	34	8	1	0.39705882352941174	0.6854647233823904	0.16401489074257258
772	111	31	8	1	0.4475806451612903	0.6849966189342125	0.16431157213069608
773	113	29	8	1	0.4870689655172414	0.688608956187972	0.16202733307235542
776	115	27	8	1	0.5324074074074074	1.0	-0.0
777	119	23	8	1	0.6467391304347826	1.0	-0.0
778	120	22	8	1	0.6818181818181818	1.0	-0.0
779	121	21	8	1	0.7202380952380952	1.0	-0.0
780	122	20	8	1	0.7625	1.0	-0.0
781	125	17	8	1	0.9191176470588235	1.0	-0.0
782	127	15	8	1	1.0583333333333333	1.0	-0.0
785	128	14	8	1	1.1428571428571428	1.0	-0.0
786	130	12	8	1	1.3541666666666667	0.5655253555991285	0.24754791850170255
787	131	11	9	0	0.0	1.0	-0.0
788	133	9	9	0	0.0	1.0	-0.0
789	134	8	9	0	0.0	1.0	-0.0
793	138	4	9	0	0.0	1.0	-0.0
797	139	3	9	0	0.0	1.0	-0.0
799	140	2	9	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

1	3
73	74

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf00047	Immunoglobulin domain	151	147	0	4	0.0
auto_disruption_pf07679	Immunoglobulin I-set domain	151	147	2	2	0.0999999999999999

domain_retention tables

frame_domain_contingency_table

73	69
1	8

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein tyrosine and serine/threonine kinase	151	142	3	6	0.22663457681508226

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
10	145	152	0.9539473684210527

frequency tables

Partner_gene_counts

Partner_gene	Event_count
BABAM2	1
CCDC149	2
CTBP1	1
DGKQ	1

Partner_gene	Event_count
GRK4	1
JAKMIP1	2
LPCAT2	2
MAEA	1
NSD2	3
PTBP1	1
RBM20	1
ROBO1	1
SLIT2	1
TACC3	131
TNIP2	3

Per-event results (results.tsv)

event_id	sampl e_id	pa tie nt _i d	fusion _nam e	part ner _ge ne	fra me _sta tus	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain _statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site2 _effect_on_f rame
EVT-P-000 0025-T03-I M6-1	P-0000 025-T0 3-IM6		DGKQ -FGF R3	DG KQ	unk now n	thre e_ _pr ime	179570 1	1	14	False	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase; Immun oglobu lin do main; I mmun oglobu linl-set domai n			DGKQ (NM_001347) - FGFR3 (NM_000142) rearrangement: c.352-750: DGKQ_c.40:FGFR3inv Protein Fusion: mid-exon {DGKQ:FGFR3}	NA
EVT-P-002 0773-T01-I M6-2	P-0020 773-T0 1-IM6		FGFR 3-BAB AM2	BAB AM2	unk now n	thre e_ _pr ime	180310 5	4	153	False	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase; Immun oglobu linl-set domai n	Immu noglo bulin doma in		BRE (NM_004899) - FGFR3 (NM_000142) rearrangement: t(2;4)(p23. 2;p16.3)(chr2:g.28322454: :chr4:g.1803105) Protein Fusion: mid-exon {BRE:FGFR3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0054353-T02-IM7-4	P-0054353-T02-IM7		FGFR3-CCDC149	CCDC149	in-frame	five_prime	1808701	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - CCDC149 (NM_001130726) fusion: c.2274+40:FGFR3_c.490-99:CCDC149inv Protein Fusion: in frame {FGFR3:CCDC149}	NA
EVT-P-0054353-T01-IM6-5	P-0054353-T01-IM6		FGFR3-CCDC149	CCDC149	in-frame	five_prime	1808701	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - CCDC149 (NM_001130726) fusion: c.2274+40:FGFR3_c.490-99:CCDC149inv Protein Fusion: in frame {FGFR3:CCDC149}	NA
EVT-P-0057559-T02-IM7-6	P-0057559-T02-IM7		FGFR3-CTBP1	CTBP1	in-frame	five_prime	1808812	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - CTBP1 (NM_001328) fusion: c.2275-31:FGFR3_c.196-1497:CTBP1inv Protein Fusion: in frame {FGFR3:CTBP1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 9022-T02-IM6-35	P-0019 022-T02-IM6		FGFR3-GRK4	GRK4	unknown	threonine_prime	1795705	1	15	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			GRK4 (NM_182982) - FGFR3 (NM_000142) rearrangement: c.339+319 1:GRK4_c.44:FGFR3dup Protein Fusion: mid-exon {GRK4:FGFR3}	NA
EVT-P-000 2610-T01-IM3-36	P-0002 610-T01-IM3		FGFR3-JAKMIP1	JAKMIP1	in-frame	five_prime	1808734	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			NA Protein fusion: in frame (FGFR3-JAKMIP1)	NA
EVT-P-003 4419-T01-IM6-37	P-0034 419-T01-IM6		FGFR3-JAKMIP1	JAKMIP1	in-frame	five_prime	1808749	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - JAKMIP1 (NM_001099433) fusion (FGFR3 exons 1-17 fused to JAKMIP1 exons 4-21): c.2274+88:FGFR3_c.624+9161:JAKMIP1inv Protein Fusion: in frame {FGFR3:JAKMIP1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0068474-T01-IM7-39	P-0068474-T01-IM7		FGFR3-MAEA	MAEA	unknown	five_prime	1808895	17	776	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-like set domain			FGFR3 (NM_000142) - MAEA (NM_001017405) fusion: c.2327:FGFR3_c.457-2460:MAEAdup Protein Fusion: mid-exon {FGFR3:MAEA}	NA
EVT-P-0010465-T01-IM5-42	P-0010465-T01-IM5		FGFR3-NSD2	NSD2	unknown	five_prime	1807917	13	653	True	disrupted	0.6570397111913358	True	Immunoglobulin domain; Immunoglobulin-like set domain		Protein tyrosine and serine/threonine kinase	FGFR3 (NM_000142) - WHSC1 (NM_001042424) rearrangement:c.1959+17:FGFR3_c.-29-12740:WHSC1del Transcript fusion (FGFR3-WHSC1)	NA
EVT-P-0013954-T01-IM5-43	P-0013954-T01-IM5		FGFR3-NSD2	NSD2	unknown	three_prime	1808929	17	787	False	lost	0.0	False		Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-like set domain		WHSC1 (NM_001042424) - FGFR3 (NM_000142) rearrangement: c.3255+4:WHSC1_c.2361:FGFR3dup Protein fusion: mid-exon (WHSC1-FGFR3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0024991-T01-IM6-44	P-0024991-T01-IM6		FGFR3-ROBO1	ROBO1	unknown	five_prime	1808963	17	799	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - ROBO1 (NM_002941) fusion: t(3;4)(p12.3;p16.3)(chr3:g.78737098::chr4:g.1808963) Protein Fusion: mid-exon {FGFR3:ROBO1}	NA
EVT-P-0022387-T02-IM6-48	P-0022387-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808707	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+46:FGFR3_c.1942-224:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0044466-T02-IM6-49	P-0044466-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808773	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-70:FGFR3_c.1645-159:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0055884-T01-IM6-50	P-0055884-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808886	17	773	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2318:FGFR3_c.1942-928:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0000941-T02-IM6-51	P-0000941-T02-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808894	17	776	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2326:FGFR3_c.1942-988:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0056846-T01-IM6-52	P-0056846-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808698	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+37:FGFR3_c.1942-426:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0053694-T03-IM6-53	P-0053694-T03-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808911	17	781	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2343:FGFR3_c.1942-151:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0053694-T02-IM6-54	P-0053694-T02-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808911	17	781	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2343:FGFR3_c.1942-151:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0016989-T02-IM6-55	P-0016989-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808718	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+57:FGFR3_c.1645-139:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0053694-T01-IM6-57	P-0053694-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808911	17	781	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2343:FGFR3_c.1942-151:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0052352-T02-IM6-59	P-0052352-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808764	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-79:FGFR3_c.1837-58:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0052324-T01-IM6-60	P-0052324-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808803	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-40:FGFR3_c.1942-486:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0059195-T01-IM7-61	P-0059195-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808841	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-2:FGFR3_c.1645-51:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0060831-T01-IM7-62	P-0060831-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808827	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-16:FGFR3_c.1941+414:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0060879-T01-IM7-63	P-0060879-T01-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808945	17	793	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2377:FGFR3_c.1942-241:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 1503-T01-IM5-64	P-0011 503-T01-IM5		FGFR3-TACC3	TACC3	unknown	five_prime	1808967	17	800	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 with TACC3 exons 10-16): c.2399:FGFR3_c.1836+75:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-005 1755-T01-IM6-65	P-0051 755-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808806	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-37:FGFR3_c.1942-333:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-001 0534-T01-IM5-66	P-0010 534-T01-IM5		FGFR3-TACC3	TACC3	unknown	five_prime	1808859	17	764	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 10-16): c.2363:FGFR3_c.1784:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0010096-T03-IM6-67	P-0010096-T03-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808925	17	786	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2357:FGFR3_c.1941+378:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0060879-T02-IM7-68	P-0060879-T02-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808945	17	793	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2377:FGFR3_c.1942-241:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0062326-T01-IM7-69	P-0062326-T01-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808913	17	782	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2345:FGFR3_c.1386-12:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0051602-T01-IM6-70	P-0051602-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808884	17	772	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2316:FGFR3_c.1942-997:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0051575-T01-IM6-71	P-0051575-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808724	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+63:FGFR3_c.1461+64:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0051267-T01-IM6-72	P-0051267-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808672	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+11:FGFR3_c.1645-143:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005 1173-T02-IM6-73	P-0051 173-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808716	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+55:FGFR3_c.1941+61:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-006 3306-T01-IM7-74	P-0063 306-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808764	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+55:FGFR3_c.1461+18:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-006 3929-T01-IM7-75	P-0063 929-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808782	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-61:FGFR3_c.1941+145:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0051724-T01-IM6-76	P-0051724-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808760	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-83:FGFR3_c.1942-112:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0050757-T02-IM7-77	P-0050757-T02-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808660	16	758	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2273:FGFR3_c.1572:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0006879-T01-IM5-78	P-0006879-T01-IM5		FGFR3-TACC3	TACC3	unknown	five_prime	1808967	17	800	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) Fusion (FGFR3 exons 1 to 18 fused with TACC3 exons 11-16) : c.2399:FGFR3_c.1941+196:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025859-T01-IM6-79	P-0025859-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1800997	2	42	False	lost	0.0	False		Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin I-set domain		FGFR3 (NM_000142) - TACC3 (NM_006342) rearrangement: c.126:FGFR3_c.1941+33:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0050136-T01-IM6-80	P-0050136-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808922	17	785	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2354:FGFR3_c.2040:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0027054-T03-IM6-81	P-0027054-T03-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808898	17	777	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2330:FGFR3_c.1942-902:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027054-T05-IM6-82	P-0027054-T05-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808898	17	777	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2330:FGFR3_c.1942-902:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0027054-T07-IM7-83	P-0027054-T07-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808898	17	777	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2330:FGFR3_c.1942-902:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0049741-T02-IM6-84	P-0049741-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808830	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-13:FGFR3_c.1942-479:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049502-T01-IM6-85	P-0049502-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808726	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+65:FGFR3_c.1941+538:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0048985-T01-IM6-86	P-0048985-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808827	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-16:FGFR3_c.1788:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0048947-T01-IM6-87	P-0048947-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808898	17	777	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2330:FGFR3_c.1941+104:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006660-T02-IM6-88	P-0006660-T02-IM6		FGFR3-TACC3	TACC3	out-of-frame	five_prime	1808678	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+17:FGFR3_c.1591+1556:TACC3dup Protein Fusion: out of frame {FGFR3:TACC3}	NA
EVT-P-0047604-T01-IM6-89	P-0047604-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808692	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+31:FGFR3_c.1941+484:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0045507-T01-IM6-90	P-0045507-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808733	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+72:FGFR3_c.1942-413:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0045382-T01-IM6-91	P-0045382-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808903	17	779	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2335:FGFR3_c.1942-101:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0047573-T01-IM6-92	P-0047573-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808844	17	759	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2276:FGFR3_c.1942-892:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0064650-T01-IM7-93	P-0064650-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808781	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-62:FGFR3_c.1837-81:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006 5466-T01-IM7-94	P-0065 466-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808795	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-like set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-48:FGFR3_c.1942-685:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-000 5361-T01-IM5-96	P-0005 361-T01-IM5		FGFR3-TACC3	TACC3	unknown	five_prime	1803212	4	188	False	lost	0.0	False	Immunoglobulin domain	Protein tyrosine and serine/threonine kinase	Immunoglobulin-like set domain	FGFR3 (NM_000142) -TACC3 (NM_006342) Fusion (FGFR3 exon 5 fused to TACC3 exon 14) : c.564:FGFR3_c.2223+249:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-000 5128-T01-IM5-97	P-0005 128-T01-IM5		FGFR3-TACC3	TACC3	in-frame	five_prime	1808814	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-like set domain			FGFR3(NM_000142) -TACC3 (NM_006342) Fusion (FGFR3 exon 17 fused to TACC3 exon 11) : c.2275-29:FGFR3_c.1942-314:TACC3dup Protein fusion: in frame (FGFR3-TACC3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 2266-T01-IM6-98	P-0032 266-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808745	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+84:FGFR3_c.1645-165:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-001 6238-T03-IM7-99	P-0016 238-T03-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808706	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+84:FGFR3_c.1942-654:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-004 7604-T02-IM6-100	P-0047 604-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808692	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+31:FGFR3_c.1941+484:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006 6401-T01-IM7-101	P-0066 401-T01-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808930	17	788	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2362:FGFR3_c.1942-935:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-000 4694-T01-IM5-102	P-0004 694-T01-IM5		FGFR3-TACC3	TACC3	unknown	five_prime	1808881	17	771	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3(NM_000142) -TACC3(NM_006342) Fusion (FGFR3 exon 17 fused with TACC3 exon 9) : c.2313:FGFR3_c.1837-88:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-000 3853-T01-IM5-103	P-0003 853-T01-IM5		FGFR3-TACC3	TACC3	in-frame	five_prime	1808680	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1 to 17 with TACC3 exons 11 to 16) Protein fusion: in frame (FGFR3-TACC3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0003827-T01-IM5-104	P-0003827-T01-IM5		FGFR3-TACC3	TACC3	in-frame	five_prime	1808714	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 11-16): c.1941+232:TACC3_c.2274+53:FGFR3dup Protein fusion: in frame (FGFR3-TACC3)	NA
EVT-P-0046580-T03-IM6-105	P-0046580-T03-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808716	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+55:FGFR3_c.1943:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0046580-T02-IM6-106	P-0046580-T02-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808716	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+55:FGFR3_c.1943:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0045507-T04-IM7-107	P-0045507-T04-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808733	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+72:FGFR3_c.1942-413:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0037126-T01-IM6-108	P-0037126-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808878	17	770	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2310:FGFR3_c.1837-16:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0038253-T01-IM6-109	P-0038253-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808852	17	762	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) -TACC3(NM_006342) Fusion (FGFR3 exon18 fused with TACC3 exon8) : c.2284:FGFR3_c.1644+106:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0038318-T01-IM6-110	P-0038318-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808756	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-87:FGFR3_c.1462-57:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0038414-T01-IM6-111	P-0038414-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808869	17	767	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2301:FGFR3_c.1941+974:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0039109-T01-IM6-112	P-0039109-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808760	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-83:FGFR3_c.162+65:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0039109-T02-IM6-113	P-0039109-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808760	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-83:FGFR3_c.162+65:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0039116-T01-IM6-114	P-0039116-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808698	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+37:FGFR3_c.944:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0039138-T02-IM6-115	P-0039138-T02-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808645	16	753	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2258:FGFR3_c.1941+335:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 9166-T01-IM6-116	P-0039 166-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808883	17	772	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2315:FGFR3_c.1942-61:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-006 7303-T01-IM7-117	P-0067 303-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808812	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-31:FGFR3_c.1941+863:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-003 9655-T01-IM6-118	P-0039 655-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808887	17	773	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2319:FGFR3_c.1836+86:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0003279-T02-IM6-119	P-0003279-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808708	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) rearrangement: c.2274+47:FGFR3_c.1941+850:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0040543-T01-IM6-120	P-0040543-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808696	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+35:FGFR3_c.1836+39:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0067315-T01-IM7-121	P-0067315-T01-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808877	17	770	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2309:FGFR3_c.1941+162:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0067602-T01-IM7-122	P-0067602-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808775	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-68:FGFR3_c.1836+111:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0047905-T01-IM6-123	P-0047905-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808802	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-41:FGFR3_c.1837-96:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0042157-T06-IM7-124	P-0042157-T06-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808855	17	763	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2287:FGFR3_c.1942-923:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 2393-T01-IM6-125	P-0042 393-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808709	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+48:FGFR3_c.1644+132:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-004 2515-T01-IM6-126	P-0042 515-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808841	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-2:FGFR3_c.1942-418:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-004 3076-T01-IM6-127	P-0043 076-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808926	17	786	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2358:FGFR3_c.2019-34:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 4057-T01-IM6-128	P-0044 057-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808714	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+53:FGFR3_c.1645-189:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-001 6624-T01-IM6-130	P-0016 624-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808733	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exon 17 fused to TACC3 exon 5): c.2274+72:FGFR3_c.1387:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-004 5507-T03-IM7-131	P-0045 507-T03-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808733	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+72:FGFR3_c.1942-413:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0054391-T01-IM6-132	P-0054391-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808789	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-54:FGFR3_c.1941+447:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0006429-T01-IM5-133	P-0006429-T01-IM5		FGFR3-TNIP2	TNIP2	unknown	five_prime	1808801	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TNIP2 (NM_024309) rearrangement: c.2275-42:FGFR3_c.147:TNIP2inv Protein fusion: mid-exon (FGFR3-TNIP2)	NA
EVT-P-0006429-T02-IM6-134	P-0006429-T02-IM6		FGFR3-TNIP2	TNIP2	unknown	five_prime	1808753	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TNIP2 (NM_024309) fusion: c.2275-90:FGFR3_c.195:TNIP2inv Protein Fusion: mid-exon {FGFR3:TNIP2}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0050971-T01-IM6-135	P-0050971-T01-IM6		FGFR3-TNIP2	TNIP2	unknown	five_prime	1808827	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TNIP2 (NM_024309) fusion: c.2275-16:FGFR3_c.252:TNIP2inv Protein Fusion: mid-exon {FGFR3:TNIP2}	NA
EVT-P-0012257-T01-IM5-136	P-0012257-T01-IM5		LPCAT2-FGFR3	LPCAT2	in-frame	five_prime	1808759	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - LPCAT2 (NM_017839) rearrangement: t(4;16)(p16.3;q12.2)(chr4:g.1808759::chr16:g.55556795) Protein fusion: in frame {FGFR3-LPCAT2}	NA
EVT-P-0012257-T03-IM6-137	P-0012257-T03-IM6		LPCAT2-FGFR3	LPCAT2	in-frame	five_prime	1808708	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - LPCAT2 (NM_017839) rearrangement: t(4;16)(p16.3;q12.2)(chr4:g.1808708::chr16:g.55556796) Protein Fusion: in frame {FGFR3-LPCAT2}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 4146-T01-I M6-139	P-0024 146-T0 1-IM6		NSD2- FGFR 3	NSD 2	in-frame	threonine _prime	180141 4	3	127	True	retained	1.0	False	Protein tyrosine and serine/ threonine kinase; Immunoglobulin-set domain	Immunoglobulin domain		WHSC1 (NM_001042424) - FGFR3 (NM_000142) rearrangement: c.761-178:WHSC1_c.380-60:FGFR3 dup Protein Fusion: in frame {WHSC1:FGFR3}	NA
EVT-P-001 7789-T01-I M6-141	P-0017 789-T0 1-IM6		PTBP 1-FGF R3	PTB P1	in-frame	five _prime	180867 6	16	758	True	retained	1.0	False	Protein tyrosine and serine/ threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - PTBP1 (NM_002819) rearrangement: t(4;19)(p16.3;p13.3)(chr4:g.1808676::chr19:g.802509) Protein Fusion: in frame {FGFR3:PTBP1}	NA
EVT-P-003 3176-T01-I M6-142	P-0033 176-T0 1-IM6		RBM2 0-FGF R3	RB M20	unknown	five _prime	180316 8	4	174	False	lost	0.0	False	Immunoglobulin domain	Protein tyrosine and serine/ threonine kinase	Immunoglobulin domain	FGFR3 (NM_000142) - RBM20 (NM_001134363) rearrangement: t(4;10)(p16.3;q25.2)(chr4:g.1803168::chr10:g.112544434) Protein Fusion: mid-exon {FGFR3:RBM20}	NA
EVT-P-003 6563-T01-I M6-143	P-0036 563-T0 1-IM6		TACC 3-FGF R3	TAC C3	out-of-frame	five _prime	180489 0	7	311	True	lost	0.0	False	Immunoglobulin domain; Immunoglobulin-set domain	Protein tyrosine and serine/ threonine kinase		FGFR3 (NM_000142) - TACC3 (NM_006342) Fusion(FGFR3 exon7 fused with TACC3 exon4) : c.931-529:FGFR3_c.305+1845:TACC3dup Protein Fusion: out of frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007131-T01-IM5-144	P-0007131-T01-IM5		TACC3-FGFR3	TACC3	in-frame	five_prime	1808764	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 with TACC3 exons 10-16): c.2275-79:FGFR3_c.1837-97:TACC3dup Protein fusion: in frame (FGFR3-TACC3)	NA
EVT-P-0028647-T01-IM6-145	P-0028647-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808731	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 11-16): c.2274+70:FGFR3_c.1942-150:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0030871-T01-IM6-146	P-0030871-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808856	17	763	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2288:FGFR3_c.1608:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000 0289-T01-I M3-147	P-0000 289-T01-IM3		TACC3-FGFR3	TACC3	unknown	five_prime	1806109	8	376	False	lost	0.0	False	Immunoglobulin domain; Immunoglobulin-set domain	Protein tyrosine and serine/threonine kinase		FGFR3 (NM_000142) TACC3 (NM_006342) Duplication (c.1128_c.1592-1871dup) Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-003 2432-T01-I M6-148	P-0032 432-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808674	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+13:FGFR3_c.1837-134:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-002 8321-T01-I M6-149	P-0028 321-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1807884	13	648	False	disrupted	0.6389891696750902	True	Immunoglobulin domain; Immunoglobulin-set domain		Protein tyrosine and serine/threonine kinase	FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-14 fused to TACC3 exons 7-16): c.1943:FGFR3_c.1591+377:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-002 7211-T01-I M6-151	P-0027 211-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808735	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 10-16): c.2274+74:FGFR3_c.1836+111:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027183-T02-IM6-152	P-0027183-T02-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808673	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) Fusion (FGFR1 exons 1-17 fused with TACC3 exons 11-16) : c.2274+12:FGFR3_c.1942-636:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0031155-T01-IM6-153	P-0031155-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808715	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused with TACC3 exons 8-16. This includes the kinase domain of FGFR3.): c.2274+54:FGFR3_c.1644+125:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0026783-T01-IM6-154	P-0026783-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808848	17	760	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 with TACC3 exons 11-16) : c.2280:FGFR3_c.1942-78:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0024954-T02-IM6-155	P-0024954-T02-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808755	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 11-16): c.2275-88:FGFR3_c.1942-76:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0024954-T01-IM6-156	P-0024954-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808755	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 11-16): c.2275-88:FGFR3_c.1942-76:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0002351-T01-IM3-157	P-0002351-T01-IM3		TACC3-FGFR3	TACC3	unknown	five_prime	1808871	17	768	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) and TACC3 (NM_006342) duplication (71843bp): FGFR3:c.2303_TACC3:c.1620dup Protein fusion: mid-exon (FGFR3-TACC3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0024346-T01-IM6-158	P-0024346-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808859	17	764	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 6-16): c.2291:FGFR3_c.1461+69:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0033514-T01-IM6-159	P-0033514-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808883	17	772	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 4-16): c.2315:FGFR3_c.927:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0023261-T01-IM6-160	P-0023261-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808927	17	787	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to exons 11-16 of TACC3): c.2359:FGFR3_c.1941+673:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 2569-T01-I M6-161	P-0022 569-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808705	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused with TACC3 exons 8-16. This includes the kinase domain of FGFR3.): c.2274+44:FGFR3_c.1645-194:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-002 0355-T01-I M6-162	P-0020 355-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808807	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 11-16): c.2275-36:FGFR3_c.1942-312:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-001 9730-T01-I M6-163	P-0019 730-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808758	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused in-frame with TACC3 exons 10-16):c.2275-85:FGFR3_c.1836+25:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0034183-T01-IM6-164	P-0034183-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808914	17	782	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-like set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 7-16): c.2346:FGFR3_c.1591+1975dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0017551-T01-IM5-165	P-0017551-T01-IM5		TACC3-FGFR3	TACC3	unknown	five_prime	1808877	17	770	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-like set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused with TACC3 exons 10-16): c.2309:FGFR3_c.1836+8:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0017444-T01-IM6-166	P-0017444-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808864	17	766	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-like set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused with TACC3 exons 11-16): c.2296:FGFR3_c.1941+70:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0017056-T01-IM6-167	P-0017056-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808671	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused in-frame with TACC3 exons 11-16): c.2274+10:FGFR3_c.1941+881:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0016989-T01-IM6-168	P-0016989-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808718	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 8-16): c.2274+57:FGFR3_c.1645-139:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0001453-T01-IM3-169	P-0001453-T01-IM3		TACC3-FGFR3	TACC3	unknown	five_prime	1808946	17	793	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			NA Protein fusion: mid-exon (FGFR3-TACC3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0001453-T01-IM3-170	P-0001453-T01-IM3		TACC3-FGFR3	TACC3	unknown	five_prime	1808946	17	793	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			NA Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0016238-T01-IM6-171	P-0016238-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808706	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exon 17 fused to TACC3 exon 11): c.2274+45:FGFR3_c.1942-654:TACC3inv Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0015863-T01-IM6-173	P-0015863-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808880	17	771	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 8-16): c.2312:FGFR3_c.1644+147:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015195-T01-IM6-174	P-0015195-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808757	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exon17 fused to TACC3 exon8): c.2275-86:FGFR3_c.1645-28:TACC3dup Protein fusion: in frame (FGFR3-TACC3)	NA
EVT-P-0015024-T01-IM6-175	P-0015024-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808210	15	677	True	disrupted	0.7436823104693141	True	Immunoglobulin domain; Immunoglobulin-set domain		Protein tyrosine and serine/threonine kinase	FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exon 16 fused to TACC3 exon 5) : c.2031-63:FGFR3_c.1385+14:TACC3dup Protein fusion: in frame (FGFR3-TACC3)	NA
EVT-P-0014530-T01-IM6-176	P-0014530-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808715	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) Fusion (FGFR3 exons 1-17 fused with TACC3 exons 6-16) : c.2274+54:FGFR3_c.1462-52:TACC3dup Protein fusion: in frame (FGFR3-TACC3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014237-T01-IM6-177	P-0014237-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808958	17	797	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 with TACC3 exons 10-16): c.2390:FGFR3_c.1836+80:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0012654-T01-IM5-178	P-0012654-T01-IM5		TACC3-FGFR3	TACC3	unknown	five_prime	1808935	17	789	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) -TACC3 (NM_006342)fusion: c.2367:FGFR3_c.1645-121:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0012373-T01-IM5-179	P-0012373-T01-IM5		TACC3-FGFR3	TACC3	in-frame	five_prime	1808730	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 with TACC3 exons 10-16): c.2274+69:FGFR3_c.1837-70:TACC3dup Protein fusion: in frame (FGFR3-TACC3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 4666-T01-I M6-180	P-0034 666-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808721	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+60:FGFR3_c.1941+226:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-003 6305-T01-I M6-181	P-0036 305-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808902	17	778	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 - (NM_000142) TACC3 (NM_006342) Fusion (FGFR3 exon18 fused with TACC3 exon3) : c.2334:FGFR3_c.274:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-000 8649-T01-I M5-182	P-0008 649-T01-IM5		TACC3-FGFR3	TACC3	unknown	five_prime	1808878	17	770	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (exons 1-18 of FGFR3 fused to exons 7-16 of TACC3); c.2310:FGFR3_c.1645-1546:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 9600-T01-IM6-183	P-0029 600-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808697	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+36:FGFR3_c.1644+158:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-000 7082-T03-IM6-184	P-0007 082-T03-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808872	17	768	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3(NM_000142) - TACC3(NM_006342) Fusion (FGFR3 exon 18 fused to TACC3 exon 8) : c.2304:FGFR3_c.1644+118:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-000 7082-T02-IM6-185	P-0007 082-T02-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808872	17	768	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (exons 1-18 of FGFR3 fused to exons 8-16 of TACC3): c.2304:FGFR3_1644+118:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

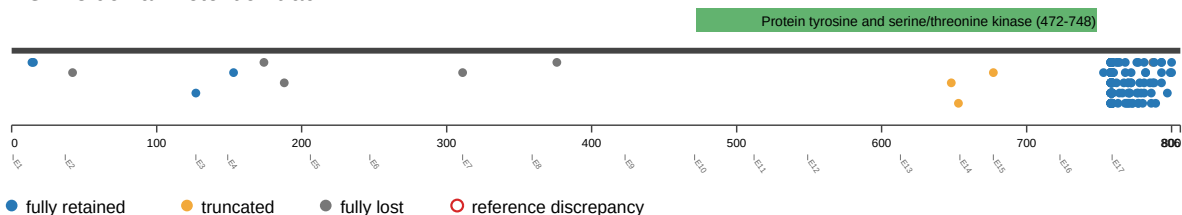
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005966-T01-IM5-186	P-0005966-T01-IM5		TACC3-FGFR3	TACC3	unknown	five_prime	1808853	17	762	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1 to 17 with TACC3 exons 11 to 16) : c.2285:FGFR3_c.1941+423:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0005862-T01-IM5-187	P-0005862-T01-IM5		TACC3-FGFR3	TACC3	in-frame	five_prime	1808707	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (exons 1-17 of FGFR3 fused to exons 14-16 of TACC3): c.2274+46:FGFR3_c.2223+896:TACC3dup. Protein fusion: in frame (FGFR3-TACC3)	NA
EVT-P-0004902-T01-IM5-189	P-0004902-T01-IM5		TACC3-FGFR3	TACC3	unknown	five_prime	1808906	17	780	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 with TACC3 exons 11-16) : c.2338:FGFR3_c.1941+300TACC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0003677-T01-IM5-190	P-0003677-T01-IM5		TACC3-FGFR3	TACC3	unknown	five_prime	1808931	17	788	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			c.2363:FGFR3_c.1784:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0003097-T01-IM5-191	P-0003097-T01-IM5		TACC3-FGFR3	TACC3	in-frame	five_prime	1808841	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			NA Protein fusion: in frame (FGFR3-TACC3)	NA
EVT-P-0006396-T01-IM6-192	P-0036396-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808747	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+86:FGFR3_c.162+46:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 6735-T01-I M6-193	P-0016 735-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808879	17	771	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 8-16); c.2311:FGFR3_c.1644+71:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-000 2862-T01-I M3-194	P-0002 862-T01-IM3		TACC3-FGFR3	TACC3	unknown	five_prime	1808873	17	769	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			NA Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-002 4463-T01-I M6-195	P-0024 463-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808801	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin-set domain			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused in-frame with TACC3 exons 10-16);c.2275-42:FGFR3_c.1836+11:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

domain retention outliers

FGFR3 domain-retention track



fusion schematic

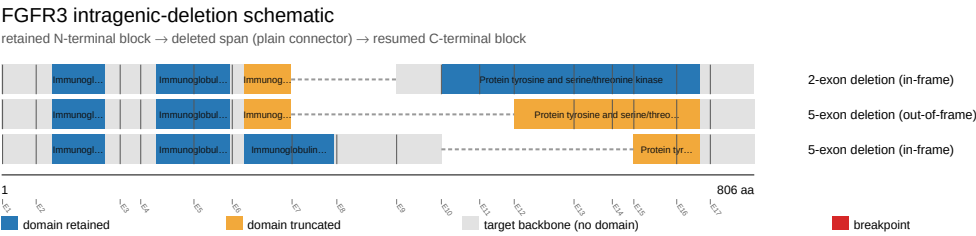
FGFR3 fusion-transcript schematic

partner block → breakpoint → retained FGFR3 portion (domain-colored, exon ticks)



Showing the top 28 of 52 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic



reference comparison

Reference vs msk_impact_50k_2026

